

$$\neg(A \wedge B) \vdash \neg A \vee \neg B.$$

1	$\neg(A \wedge B)$	
2	$\neg(\neg A \vee \neg B)$	hip. ad
3	$\neg A$	hip. ad
4	$\neg A \vee \neg B$	iv 3
5	$\neg \neg A$	i 2, 3, 4
6	A	e 7 5
7	$\neg B$	hip. ad
8	$\neg A \vee \neg B$	iv 7
9	$\neg \neg B$	i 2, 7, 8
10	B	e 7 9
11	$A \wedge B$	i 6, 10
12	$\neg \neg(\neg A \vee \neg B)$	i 1, 2, 11
13	$\neg A \vee \neg B$	e 12

$$\neg A \vee \neg B \vdash \neg(A \wedge B).$$

1	$\neg A \vee \neg B$	
2	$\neg A$	hip. ad
3	$A \wedge B$	hip. ad
4	A	e 1 3
5	$\neg(A \wedge B)$	i 2, 3, 4
6	$\neg B$	hip. ad
7	$A \wedge B$	hip. ad
8	B	e 1 7
9	$\neg(A \wedge B)$	i 6, 7, 8
10	$\neg(A \wedge B)$	e 1-9

$$\neg(A \vee B) \vdash \neg A \wedge \neg B.$$

1	$\neg(A \vee B)$	
2	A	hip. ad
3	$A \vee B$	iv 2
4	$\neg A$	i 1, 2, 3
5	B	hip. ad
6	$A \vee B$	iv 5
7	$\neg B$	i 1, 5, 6
8	$\neg A \wedge \neg B$	i 4, 7

1	$\neg A \wedge \neg B$	
2	$\neg A$	e 1
3	$\neg B$	e 1
4	$A \vee B$	hip. ad
5	A	hip. ad
6	$\neg(\neg A \wedge \neg B)$	i 1, 2, 3
7	B	hip. ad
8	$\neg(\neg A \wedge \neg B)$	i 1, 3, 7
9	$\neg(\neg A \wedge \neg B)$	e 4, 5-6, 7-8
10	$\neg(A \vee B)$	i 1, 4, 9

$$A \vee (B \wedge C) \vdash (A \vee B) \wedge (A \vee C).$$

1	$A \vee (B \wedge C)$	
2	A	hip. ad
3	$A \vee B$	iv 2
4	$A \vee C$	iv 2
5	$(A \vee B) \wedge (A \vee C)$	i 3, 4
6	$B \wedge C$	hip. ad
7	B	e 1 6
8	C	e 1 6
9	$A \vee B$	iv 7
10	$A \vee C$	iv 8
11	$(A \vee B) \wedge (A \vee C)$	i 9, 10
12	$(A \vee B) \wedge (A \vee C)$	e 1, 2-5, 6-11

$$(A \vee B) \wedge (A \vee C) \vdash A \vee (B \wedge C).$$

1	$(A \vee B) \wedge (A \vee C)$	
2	$A \vee B$	e 1
3	$A \vee C$	e 1
4	A	hip. ad
5	$A \vee (B \wedge C)$	iv 4
6	B	hip. ad
7	A	hip. ad
8	$A \vee (B \wedge C)$	iv 7
9	C	hip. ad
10	$B \wedge C$	i 6, 9
11	$A \vee (B \wedge C)$	iv 10
12	$A \vee (B \wedge C)$	e 1-3
13	$A \vee (B \wedge C)$	e 12

$$A \wedge (B \vee C) \vdash (A \wedge B) \vee (A \wedge C).$$

1	$A \wedge (B \vee C)$	
2	A	e 1
3	$B \vee C$	e 1
4	B	hip. ad
5	$A \wedge B$	i 2, 4
6	$(A \wedge B) \vee (A \wedge C)$	iv 5
7	C	hip. ad
8	$A \wedge C$	i 2, 7
9	$(A \wedge B) \vee (A \wedge C)$	iv 8
10	$(A \wedge B) \vee (A \wedge C)$	e 1, 2-6, 7-9

$$(A \wedge B) \vee (A \wedge C) \vdash A \wedge (B \vee C).$$

1	$(A \wedge B) \vee (A \wedge C)$	
2	$A \wedge B$	hip. ad
3	A	e 2
4	B	e 2
5	$B \vee C$	iv 4
6	$A \wedge (B \vee C)$	i 3, 5
7	$A \wedge C$	hip. ad
8	A	e 7
9	C	e 7
10	$B \vee C$	iv 9
11	$A \wedge (B \vee C)$	i 8, 10
12	$A \wedge (B \vee C)$	e 1, 2-6, 7-11